CMAT 651 · Final Exam Problem Bank

Statistics with Python for AI · Every sample-final problem, sorted by topic

What this is. Every problem from five sample final exams, sorted into the topic it belongs to. The real final uses the same *kinds* of problems, with new numbers and new AI settings, so learn the ideas, not these answers. Every problem follows the final's rules: handwritten, no code, no calculator. Any value you need is given in the problem.

Tags. A tag such as [Judging estimators] on a part means that part belongs to a different topic than the rest of its problem. Each topic's first page lists these parts under *Also practice*.

Dig deeper. Some topics end with problems on ideas that go past the course. They are **not on the final**. They are there for presenters, and for anyone who wants more.

If you are presenting the topic

- **Work every problem in your section first,** including the tagged parts listed under *Also practice*. Submit your worked solutions with your draft. Your draft is not approved without them.
- **Prepare the class to solve every problem here.** Each kind of problem in your section (the *Presentation Guide* lists them under your topic) must be covered in your Teach, a Board problem, or your exit ticket. Your worked example must be your own, not one of these.
- **Write new Board problems.** Use the kinds of problems in your section, but invent a new AI setting and new numbers that work out by hand. Do not copy or lightly edit a problem from this bank.
- **Write your exit ticket at this level:** a concept or short derivation, a line of Python by hand, and an interpretation in an AI setting.
- **Dig.** Bring in one idea from outside the course notes: a *Dig deeper* problem, a paper, library documentation, or a real AI system. Cite it on your AI slide.

Everyone else

- Before each class, try that topic's problems. Exit tickets are written in this style, and they count toward Written Partial Review 3.
- Each topic's answer key is posted after its class.

The topics

Topic	When	Presenter	Problems	Dig deeper
Probability and Bayes' rule	Module 1	Already taught	5	–
Random variables and distributions	Module 2	Already taught	5	–
Covariance and correlation	Wed Oct 28	The group	2	1
The covariance matrix and the multivariate normal	Fri Oct 30	The group	2	–
PCA	Mon Nov 2	Ronny	2	–
The CLT and standard error	Wed Nov 4	Ty'Kuan	2	–
Concentration inequalities	Fri Nov 6	Brian	3	–
The bootstrap and the delta method	Mon Nov 9	Shirley	2	1
Maximum likelihood	Wed Nov 11	Ronny	3	–
Judging estimators	Fri Nov 13	Ty'Kuan	2	–
Bayesian inference	Mon Nov 16	Brian	5	2
Hypothesis testing	Wed Nov 18	Shirley	5	1
Linear models	Fri Nov 20	Ronny	5	–
Logistic regression	Mon Nov 23	Ty'Kuan	5	2
Honest evaluation and causal claims	Wed Nov 25	Brian	5	2

How the topics show up on a final

Each sample final is 2 hours long and has ten questions worth 15 points each, one question per area below. A question can draw on any topic in its area.

Area of the final	Topics in this bank
Probability and Bayes' rule	Probability and Bayes' rule
Random variables and distributions	Random variables and distributions
Multivariate data and PCA	Covariance and correlation; the covariance matrix and the multivariate normal; PCA
Limits and uncertainty	The CLT and standard error; concentration inequalities; the bootstrap and the delta method
Estimation	Maximum likelihood; judging estimators
Bayesian inference	Bayesian inference
Hypothesis testing	Hypothesis testing
Linear models	Linear models
Logistic regression and cross-entropy	Logistic regression
Honest evaluation	Honest evaluation and causal claims

Probability and Bayes' rule

Module 1 · already taught: review it for the final

5 problems, 75 points.

A deepfake detector and the base rate · Sample A **15 points**

On a video platform, 5% of uploads are deepfakes. A detector flags 90% of deepfakes. It also flags 5% of real videos by mistake.

(a) **(4)** Imagine 10,000 uploads. Fill in all four counts, and show how you got each one.

	Flagged	**Not flagged**
Deepfake		
Real		

(b) **(2)** A video is flagged. Find the probability that it really is a deepfake (the detector's *precision*).

(c) **(3)** Find the detector's accuracy, and the accuracy of a "detector" that never flags anything. What does the comparison tell you about judging a detector of a rare event?

(d) **(3)** A second detector flags 80% of deepfakes and 20% of real videos. Given a video's true status, the two detectors act independently. A video is flagged by *both*. Use odds to find the probability that it is a deepfake.

(e) **(3)** State exactly the assumption that part (d) relies on. Describe a realistic way it fails, and say whether your answer to (d) would then be too high or too low.

Adding up evidence in a phishing filter · Sample B **15 points**

One email in 10 is phishing. A filter uses three clues. Each likelihood ratio is $P(\text{clue} \mid \text{phishing})/P(\text{clue} \mid \text{legitimate})$.

Clue	**Likelihood ratio**
Suspicious link	6
Urgent tone	3
Sender is in the address book	$\frac{1}{2}$

(a) **(3)** Give the prior odds of phishing. An email has a suspicious link. Find the posterior odds and the probability that it is phishing.

(b) **(3)** The same email also has an urgent tone and comes from a sender in the address book. Treat the clues as independent given the class. Find the final probability. Then write the whole update as a sum of logarithms.

(c) **(2)** What does a likelihood ratio of $\frac{1}{2}$ mean? What would a likelihood ratio of 1 mean?

(d) **(4)** Prove that posterior odds = likelihood ratio × prior odds. Start from Bayes' rule, and point to the step where $P(\text{clue})$ cancels.

(e) **(3)** Name the classifier that multiplies likelihood ratios this way. State the assumption it makes, including what it is conditional on, and give a realistic way it fails here.

When the total points the wrong way · Sample C — 15 points

Two chatbots were each tested on 1,000 prompts. The table shows how many answers were correct.

	Chatbot P	Chatbot Q
Easy prompts	270 correct of 300	595 correct of 700
Hard prompts	280 correct of 700	105 correct of 300
All prompts	550 correct of 1,000	700 correct of 1,000

(a) **(3)** Find each chatbot's accuracy on easy prompts, on hard prompts and overall.

(b) **(3)** [Linear models] Which chatbot is better? Explain why the "All prompts" row says the opposite.

(c) **(4)** Real traffic is 40% easy and 60% hard. Find each chatbot's expected accuracy on real traffic. Name the rule, and write it as a formula.

(d) **(3)** An answer from chatbot P (in this test) is correct. Find the probability that its prompt was hard. Which rule did you use?

(e) **(2)** How should the two chatbots have been compared fairly?

The interview screen that invents a pattern · Sample D — 15 points

An AI resume screener invites an applicant to interview if they have a *strong coding test* or a *strong referral*, or both. Across all applicants these two are independent, and each has probability $\frac{1}{3}$.

(a) **(3)** Find $P(\text{interviewed})$. Start from the complement.

(b) **(3)** Find $P(\text{strong coding} \mid \text{interviewed})$.

(c) **(3)** Among interviewed applicants who have a strong referral, what fraction have strong coding? Compare with (b). Among interviewed applicants, are the two still independent?

(d) **(3)** Prove that if A and B are independent, then A and B^c are independent.

(e) **(3)** [Honest evaluation and causal claims] A team trains a model to predict job performance using only interviewed applicants. Name the problem, say who is missing from the data, and give one wrong conclusion the model could learn.

A radiology AI and a rare condition · Sample E — 15 points

An AI model that reads chest X-rays detects 99% of patients who have a condition (sensitivity) and wrongly flags 1% of patients who do not (false-positive rate).

(a) **(3)** In a clinic where 1% of patients have the condition, a patient is flagged. Use odds to find the probability that they have it.

(b) **(3)** Repeat for a specialist clinic where 10% of patients have the condition.

(c) **(3)** Write the probability of disease given a flag as a function of the prevalence π, the sensitivity s and the false-positive rate f. What happens as $\pi \to 0$?

(d) **(3)** In the 1% clinic, a flagged patient is re-tested with an independent second model with the same sensitivity and false-positive rate, and is flagged again. Find the probability now. State the assumption.

(e) **(3)** A vendor writes: "The false-positive rate is only 1%, so a flagged patient is 99% likely to be sick." Name the error, and correct it using (a).

Random variables and distributions

Module 2 · already taught: review it for the final

5 problems, 75 points.

Softmax turns scores into a distribution · Sample A **15 points**

A language model gives three candidate next tokens the scores (logits) $z = (\ln 9,\ \ln 4,\ 0)$ for *cat*, *dog* and *fish*. Softmax with temperature T gives $P(k) = e^{z_k/T} \big/ \sum_j e^{z_j/T}$.

(a) **(3)** Find the three probabilities at $T = 1$. Show that they form a PMF.

(b) **(3)** Find the probabilities at $T = 2$. In a sentence or two, say what raising the temperature does, and what happens as $T \to \infty$ and as $T \to 0$.

(c) **(4)** At $T = 2$, let X be the position of the sampled token in the list (1, 2 or 3). Find $E[X]$ and $\mathrm{Var}(X)$.

(d) **(3)** The sampler draws U uniform on $[0, 1)$ and returns the first token whose CDF value is greater than U. Write the CDF at $T = 2$. Which token is returned for $U = 0.40$, 0.70 and 0.90? Explain why *fish* is chosen with probability $\frac{1}{6}$.

(e) **(2)** [Logistic regression] Show that two-class softmax is the logistic sigmoid: $\dfrac{e^{z_1}}{e^{z_1} + e^{z_2}} = \dfrac{1}{1 + e^{-(z_1 - z_2)}}$.

How long does the API take to answer? · Sample B **15 points**

The response time X, in seconds, of an AI API has density $f(x) = c\,(2 - x)$ for $0 \le x \le 2$, and $f(x) = 0$ otherwise.

Values you may use. $\sqrt{2} \approx 1.41$.

(a) **(3)** Find c.

(b) **(3)** Find the CDF $F(x)$ for $0 \le x \le 2$. Use it to find $P(X \le 1)$ and $P(X > 1.5)$.

(c) **(4)** Find $E[X]$ and $\mathrm{Var}(X)$.

(d) **(3)** Find the median response time. Is it above or below the mean? What does that say about the shape of the distribution?

(e) **(2)** $f(0) = 1$. Is that the probability that a response takes 0 seconds? Explain.

Requests arriving at an LLM server · Sample C **15 points**

Requests arrive at a server as a Poisson process. The number of requests X in one second is Poisson(3), and the time T until the next request is Exponential with rate 3.

Values you may use. $e^{-3} \approx 0.05$.

(a) **(3)** Find $P(X = 0)$ and $P(X \ge 2)$.

(b) **(2)** Give the mean and variance of X. The logs show a mean of 3 per second but a variance of 12. What does that tell you?

(c) **(3)** Find $P(T > 1)$. Then show that $P(T > 1.5 \mid T > 0.5) = P(T > 1)$, and name the property.

(d) **(4)** Show that the moment generating function of Poisson(λ) is $M(t) = \exp\big(\lambda(e^t - 1)\big)$. Use it to show that requests to two independent servers, Poisson(3) and Poisson(2), add up to Poisson(5).

(e) **(3)** Name the distribution for each (half a point each): (i) whether one answer is correct; (ii) the number correct out of 50 independent questions, each correct with the same probability p; (iii) the number of sampled tokens up to and including the first stop token, if each token is the stop token independently with the same probability; (iv) the time until the next request; (v) a sum of many small independent latency delays; (vi) a prior or posterior for a click-through rate.

Counting with indicators · Sample D — 15 points

Values you may use. $\sqrt{3.2} \approx 1.79$, $0.8^4 = 0.4096$, $0.8^5 = 0.32768$.

(a) **(3)** An LLM answers 20 independent questions, each correctly with probability 0.8. Let Y be the number correct. Find $E[Y]$, $\mathrm{Var}(Y)$ and the SD.

(b) **(3)** In a demo on only 5 questions, find the probability that the model looks perfect, and the probability that it gets at least 4 right.

(c) **(4)** A deduplication step hashes 100 documents into 1,000 buckets, each document independently and uniformly. Find the expected number of pairs of documents that land in the same bucket. Why can you use linearity even though the pair events are not mutually independent?

(d) **(3)** For a single Bernoulli(p) indicator X, derive $\mathrm{Var}(X) = p(1-p)$.

(e) **(2)** Find the variance of the accuracy $Y/20$ from part (a). What is the common mistake?

Attempts until a jailbreak first works · Sample E — 15 points

A red team keeps trying a jailbreak. Each attempt succeeds with probability p, independently. Let X be the number of attempts up to and including the first success.

(a) **(3)** Write the PMF of X and show that it sums to 1.

(b) **(3)** Show that $P(X \le k) = 1 - (1-p)^k$. With $p = 0.2$, find the probability of success within 3 attempts.

(c) **(3)** Use first-step analysis ($E[X] = 1 + (1-p)E[X]$) to find $E[X]$. Explain the equation in words.

(d) **(4)** [Maximum likelihood] The first success came on attempt 3. Write the likelihood $L(p)$. Find $L(\frac{1}{3})$ and $L(\frac{1}{2})$. Which value explains the data better? Find the p that maximizes L.

(e) **(2)** Is $L(\frac{1}{3})$ the probability that $p = \frac{1}{3}$? Explain.

Covariance and correlation

Wed Oct 28 · presented by the group

2 problems, 30 points.

Do two models make the same mistakes? · Sample B **15 points**

Let $X = 1$ if model 1 gets a question wrong and $Y = 1$ if model 2 gets it wrong. On a large benchmark:

	$Y = 1$	$Y = 0$
$X = 1$	0.06	0.04
$X = 0$	0.14	0.76

(a) **(3)** Find $P(X = 1)$, $P(Y = 1)$ and $P(Y = 1 \mid X = 1)$.

(b) **(4)** Find $\mathrm{Cov}(X, Y)$ and the correlation.

(c) **(2)** If the errors were independent, how often would both models be wrong on the same question? Compare with the table, and say what the difference means.

(d) **(4)** An ensemble averages two risk scores S_1 and S_2, each with variance 4. Find the variance of the average,

$$\mathrm{Var}\left(\tfrac{S_1+S_2}{2}\right),$$

when $\mathrm{Cov}(S_1, S_2) = 1$, when the scores are independent, and when they are identical. Show the formula you used.

(e) **(2)** Use (c) and (d) to say what makes an ensemble worth building.

Two features that move together · Sample E **15 points**

Two inputs to a response-quality model, prompt length x and retrieval score y, were recorded for five queries: $x = (1, 2, 3, 4, 5)$ and $y = (2, 1, 4, 3, 5)$.

(a) **(4)** Find the sample covariance (divide by $n - 1$), both sample variances, and the correlation.

(b) **(4)** [The covariance matrix and the multivariate normal] Prove $\mathrm{Var}(aX + bY) = a^2 \mathrm{Var}\, X + b^2 \mathrm{Var}\, Y + 2ab\, \mathrm{Cov}(X, Y)$.

(c) **(4)** Use (b) with $\mathrm{Var}\left(X/\sigma_X + Y/\sigma_Y\right) \geq 0$ and $\mathrm{Var}\left(X/\sigma_X - Y/\sigma_Y\right) \geq 0$ to prove that $-1 \leq \rho \leq 1$.

(d) **(3)** If these two features enter a linear regression together, find the VIF of each $(1/(1-r^2))$, and the factor by which each coefficient's SE is inflated.

Dig deeper (not on the final: for presenters and anyone who wants more)

Ensembles and epistemic uncertainty · Dig deeper **5 points**

M models each have prediction variance σ^2 on a given input, and every pair has correlation ρ.

(a) **(3)** Show that the variance of their average is $\sigma^2\big(\rho + \frac{1-\rho}{M}\big)$. Evaluate it for $\rho = 0.5$ and $M = 5$, and as $M \to \infty$.

(b) **(2)** Why is disagreement among ensemble members used as a measure of epistemic (model) uncertainty rather than aleatoric (data) noise?

The covariance matrix and the multivariate normal

Fri Oct 30 · presented by the group

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2 problems, 30 points.

Also practice: *Two features that move together* (Covariance and correlation), part (b).

Two coordinates of an embedding · Sample C — 15 points

Two coordinates (X, Y) of a sentence embedding are bivariate normal, with $\mu_X = 10$, $\mu_Y = 20$, $\sigma_X = 1$, $\sigma_Y = 2$ and correlation $\rho = 0.6$.

(a) **(3)** Write the covariance matrix.

(b) **(3)** Find $\mathrm{Var}(X + Y)$ and $\mathrm{Var}(X - Y)$.

(c) **(4)** Find $E[Y \mid X = 12]$ and $\mathrm{Var}(Y \mid X = 12)$, using $E[Y \mid X = x] = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X)$ and $\mathrm{Var}(Y \mid X) = \sigma_Y^2(1 - \rho^2)$.

(d) **(3)** If ρ were 0, would X and Y be independent? Would the same be true for any two uncorrelated variables? Give a counterexample.

(e) **(2)** [Linear models] What is the slope $\rho\sigma_Y/\sigma_X$ in modeling terms? By what fraction does knowing X reduce the variance of Y?

Three signals in a fraud score · New problem — 15 points

A fraud model combines three signals: X_1 (how unusual the amount is), X_2 (how fast the account is spending) and X_3 (device risk). Their covariance matrix is

$$\Sigma = \begin{pmatrix} 4 & 2 & 0 \\ 2 & 9 & -1.5 \\ 0 & -1.5 & 1 \end{pmatrix}.$$

(a) **(3)** Read off $\mathrm{Var}(X_2)$ and $\mathrm{Cov}(X_1, X_2)$. Find the three correlations ρ_{12}, ρ_{23} and ρ_{13}.

(b) **(2)** Why is every covariance matrix symmetric? Why are its diagonal entries never negative?

(c) **(4)** The fraud score is $S = X_1 + X_2 + 2X_3$. Find $\mathrm{Var}(S)$.

(d) **(3)** Find $\mathrm{Var}(X_1 + X_2)$, the score without device risk. Adding $2X_3$ adds a term with variance 4, yet the score's variance goes down. Explain.

(e) **(3)** Suppose (X_1, X_2, X_3) is multivariate normal with mean vector $\mathbf{0}$. What is the distribution of S? Are X_1 and X_3 independent?

PCA

Mon Nov 2 · presented by Ronny

2 problems, 30 points.

Compressing an embedding with PCA · Sample A — 15 points

Two centered features of an image embedding have covariance matrix $\Sigma = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$.

(a) **(4)** Show that $(1, 1)$ and $(1, -1)$ are eigenvectors of Σ, and find their eigenvalues.

(b) **(3)** Give the first principal component as a unit vector. What share of the total variance does it explain? How many components do you need to keep 90%?

(c) **(4)** An embedding point is $x = (3, 1)$. Find its PC1 score, its reconstruction from PC1 alone, and the squared reconstruction error.

(d) **(2)** Find the correlation between the two features.

(e) **(2)** If one feature were measured in dollars and the other in percent, why would you standardize before running PCA?

What PCA sees in a feature set · Sample D — 15 points

(a) **(3)** Three features of a recommendation model are uncorrelated, with variances 6, 3 and 1. What are the principal components, their shares of variance, and the number needed to explain at least 90%?

(b) **(3)** Now each feature is standardized first. Redo (a). What do you learn?

(c) **(5)** A pipeline accidentally includes the same standardized feature twice, plus one unrelated feature, so the correlation matrix is $R = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Check that $(1, 1, 0)$, $(0, 0, 1)$ and $(1, -1, 0)$ are eigenvectors, find their eigenvalues, and give each component's share of variance.

(d) **(2)** [Linear models] What does the zero eigenvalue mean? What does it do to a linear regression on these three features?

(e) **(2)** A colleague says, "PC1 tells us the best feature to keep." Correct them.

The CLT and standard error

Wed Nov 4 · presented by Ty'Kuan

2 problems, 30 points.

Also practice: *How big must the test set be?* (Concentration inequalities), part (c).

Is model A really better than model B? · Sample A — 15 points

Model A scores 0.83 accuracy on 1,600 test questions. Model B scores 0.80 on a *different* 1,600 questions from the same source. Use $p = 0.8$ inside both standard errors.

Values you may use. $\sqrt{2} \approx 1.41$, $z_{0.975} = 1.96$.

(a) **(3)** Find each model's standard error and its 95% interval.

(b) **(4)** Find the standard error of the difference $\hat{p}_A - \hat{p}_B$ and a 95% interval for the true difference. Does the evidence favor A?

(c) **(2)** The two intervals in (a) overlap. A teammate says, "They overlap, so there is no difference." Explain the mistake.

(d) **(3)** Show why $\mathrm{Var}(\hat{p}_A - \hat{p}_B) = \mathrm{Var}(\hat{p}_A) + \mathrm{Var}(\hat{p}_B)$ here. Explain why this formula would be wrong if both models were scored on the *same* 1,600 questions.

(e) **(3)** You estimate model A's win rate against model B (near 0.5) by simulating N independent head-to-head matchups. How many matchups make the standard error at most 0.001?

When rows are not independent · Sample E — 15 points

Values you may use. $P(Z > 2) \approx 0.023$, $\sqrt{5.5} \approx 2.35$.

(a) **(3)** A batch has 100 independent requests. Each has latency with mean 0.2 s and SD 0.1 s. Find the mean and SD of the total latency, and use the CLT to estimate $P(\text{total} > 22 \text{ s})$.

(b) **(2)** After a run of slow requests, an engineer says the next ones are "due" to be fast, by the Law of Large Numbers. Correct them.

(c) **(5)** Let $Y_1, \ldots, Y_m$ each have variance σ^2, with every pair having correlation ρ. Show that $\mathrm{Var}(\bar{Y}) = \dfrac{\sigma^2}{m}\big(1 + (m-1)\rho\big)$.

(d) **(3)** A test set has 1,000 rows: 100 users with 10 rows each. Each row has variance σ^2, rows from the same user have correlation 0.5, and different users are independent. Find Var of the overall mean in terms of σ^2, and the effective sample size.

(e) **(2)** By what factor is the naive SE (which treats all 1,000 rows as independent) too small? What should you do instead?

Concentration inequalities

Fri Nov 6 · presented by Brian

3 problems, 45 points.

How big must the test set be? · Sample B **15 points**

A model's accuracy is measured on n independent test items, each scored 1 (right) or 0 (wrong). Let $\hat{p}$ be the measured accuracy and p the true accuracy.

Values you may use. $\ln 40 \approx 3.689$, $\ln 400 \approx 5.9915$.

(a) **(3)** State Hoeffding's inequality for $\hat{p}$. What does it assume, and what does it not need to know?

(b) **(4)** Find the smallest n that guarantees $|\hat{p} - p| < 0.02$ with probability at least 95%.

(c) **(3)** [The CLT and standard error] The CLT in the worst case ($p = 0.5$) asks for $n = (1.96 \times 0.5/0.02)^2$. Compute it. Why is Hoeffding's number larger, and why might you still use it?

(d) **(3)** The team will try 10 models and pick the best by test accuracy. To keep every one of the 10 estimates within 0.02 at once, with probability at least 95%, apply Hoeffding to each with $\delta = 0.05/10$ (the union bound). Find the new n, and say what this has to do with overfitting the test set.

(e) **(2)** A toxicity score is never negative and has mean 0.1. Bound the probability that a message scores at least 0.5.

A latency promise that must always hold · New problem **15 points**

A chatbot's response time T is never negative. Over a month, its mean is 200 ms and its standard deviation is 50 ms. The team does not know the shape of the distribution.

(a) **(2)** Use Markov's inequality to bound $P(T \geq 1000 \text{ ms})$.

(b) **(3)** Use Chebyshev's inequality to bound $P(|T - 200| \geq 150)$.

(c) **(3)** If T were Normal, the exact value in (b) would be about 0.0027. Why is Chebyshev's bound so much larger? Why might the team still quote 1/9 in a service promise?

(d) **(4)** Prove Chebyshev's inequality, $P(|T - \mu| \geq k\sigma) \leq 1/k^2$, from Markov's inequality.

(e) **(3)** The team wants a timeout that at most 4% of requests reach. Use Chebyshev to find how many standard deviations above the mean it must be, and the timeout in milliseconds.

How much can a leaderboard be trusted? · New problem **15 points**

A benchmark has $n = 5{,}000$ independent test items. A fixed model's score $\hat{p}$ is the share of items it gets right, and p is its true accuracy. Hoeffding's inequality says $P(|\hat{p} - p| \geq \varepsilon) \leq 2e^{-2n\varepsilon^2}$. Use $e^{-4} \approx 0.018$, $\ln 40 \approx 3.69$, $\ln 1000 \approx 6.91$, $\sqrt{3.69} \approx 1.92$ and $\sqrt{6.91} \approx 2.63$.

(a) **(3)** Bound $P(|\hat{p} - p| \geq 0.02)$. Say what it means in one sentence.

(b) **(4)** Find the ε that Hoeffding guarantees with probability 95%.

(c) **(4)** Twenty-five teams' models are scored on the same test set. Use the union bound to find the ε such that, with probability 95%, all 25 scores are within ε of the truth at once.

(d) **(4)** Name two assumptions behind (b). Then explain why the top score on a public leaderboard, after teams tuned their models on that same test set, is likely too high even with (c).

The bootstrap and the delta method

Mon Nov 9 · presented by Shirley

2 problems, 30 points.

The bootstrap by hand · Sample C **15 points**

An AI service's response times, in seconds, on four requests were 1, 2, 3 and 10.

(a) **(2)** Find the mean and the median.

(b) **(3)** A bootstrap resample draws 4 values with replacement. How many ordered resamples are possible? What is the probability that a resample does not contain the 10?

(c) **(3)** Find the probability that the maximum of a resample is 10. Explain why the bootstrap cannot tell you how slow the worst request could really be.

(d) **(4)** Describe, step by step, how to build a 95% percentile bootstrap interval for the median. Why sample with replacement?

(e) **(3)** In a larger study, 50 users each sent 20 requests. A colleague bootstraps the 1,000 rows. Will the interval be too wide or too narrow? What should be resampled instead?

Standard errors for transformed numbers · Sample D **15 points**

The delta method says that if $\hat{\theta}$ is approximately Normal with standard error SE, then

$$\mathrm{SE}\left(g(\hat{\theta})\right) \approx |g'(\theta)|\ \mathrm{SE}(\hat{\theta}).$$

(a) **(3)** A model hallucinates on 90 of 900 audited answers. Find $\hat{p}$ and its SE.

(b) **(4)** Logistic models work with the log-odds $g(p) = \ln \dfrac{p}{1-p}$. Find $g'(p)$ and the SE of $g(\hat{p})$.

(c) **(4)** Daily incident counts are Poisson with mean $\lambda = 4$. Over $n = 100$ days, find the SE of $\bar{X}$, and then the SE of $\sqrt{\bar{X}}$. What is special about the second answer?

(d) **(2)** Where does the delta method come from?

(e) **(2)** We used $\hat{p}$, not the unknown p, inside the SE. Which theorem says that is fine for large samples?

Dig deeper (not on the final: for presenters and anyone who wants more)

Split conformal prediction · Dig deeper **7 points**

A latency predictor's absolute errors on 19 held-out calibration requests, sorted, are 0.1, 0.2, 0.2, 0.3, 0.4, 0.4, 0.5, 0.6, 0.7, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.4, 1.6, 1.9, 2.5 seconds.

(a) **(4)** For 90% coverage, use the $\lceil (n+1)(1-\alpha) \rceil$-th smallest error as the margin. Find it, and give the interval for a new request predicted at 3.0 s.

(b) **(3)** What does the guarantee assume, and what does it *not* promise?

Maximum likelihood

Wed Nov 11 · presented by Ronny

3 problems, 45 points.

Also practice: *Time between model failures* (Judging estimators), part (a); *Time between model failures* (Judging estimators), part (c); *Attempts until a jailbreak first works* (Random variables and distributions), part (d); *Squared error is a likelihood* (Judging estimators), part (a); *Squared error is a likelihood* (Judging estimators), part (b); *Squared error is a likelihood* (Judging estimators), part (c).

The failure rate of a code-generation model · Sample A — 15 points

Each output of a code-generation model fails its unit tests with unknown probability p, independently of the other outputs. Of $n = 100$ outputs, $s = 20$ fail.

(a) **(4)** Write the likelihood $L(p)$ and the log-likelihood $\ell(p)$. Derive the MLE $\hat{p}$, check that it is a maximum, and give its value.

(b) **(4)** [Judging estimators] Find the Fisher information $I(p)$ for one output. Use it to get the standard error of $\hat{p}$ and a 95% interval.

(c) **(3)** [Judging estimators] State the Cramér–Rao bound for unbiased estimators of p. Does $\hat{p}$ reach it?

(d) **(2)** Find the MLE of the *odds* of failure. Name the property you used.

(e) **(2)** Give two reasons to maximize $\ell(p)$ rather than $L(p)$.

Counting API errors per hour · Sample B — 15 points

The number of errors an AI service throws in an hour is modeled as Poisson(λ). Five hours show 2, 0, 3, 1 and 4 errors.

Values you may use. $\sqrt{0.4} \approx 0.63$.

(a) **(4)** Derive the MLE of λ, check that it is a maximum, and compute it.

(b) **(3)** [Judging estimators] Find the Fisher information for one hour and the standard error of $\hat{\lambda}$.

(c) **(3)** A Poisson has mean λ and variance λ, so the method of moments gives two estimators. Compute both: $\bar{x}$, and the variance $\frac{1}{n}\sum(x_i - \bar{x})^2$.

(d) **(3)** In a different month the sample variance is 8 while the mean is still 2. What does that say about the model? By what factor is the standard error from (b) wrong?

(e) **(2)** [Judging estimators] Show that $\hat{\lambda}$ is unbiased.

Estimating a hidden timeout · Sample D — 15 points

A model's response times are uniform on $[0, \theta]$, where θ is an unknown timeout. Four responses took 2, 5, 8 and 3 seconds.

(a) **(3)** Find the method-of-moments estimate of θ.

(b) **(4)** Write the likelihood $L(\theta)$ for these data and find the MLE. Sketch $L(\theta)$.

(c) **(4)** [Judging estimators] Using $P(\max \le x) = (x/\theta)^n$, show that $E[\max] = \dfrac{n}{n+1}\theta$. Is the MLE biased? Give the unbiased version for these data.

(d) **(2)** For the data 1, 1, 1, 9, find the method-of-moments estimate. What is wrong with it?

(e) **(2)** [Judging estimators] Using (c), find the bias of the MLE. What happens to it as n grows? Is the MLE consistent?

Judging estimators

Fri Nov 13 · presented by Ty'Kuan

2 problems, 30 points.

Also practice: *The failure rate of a code-generation model* (Maximum likelihood), part (b); *The failure rate of a code-generation model* (Maximum likelihood), part (c); *Counting API errors per hour* (Maximum likelihood), part (b); *Counting API errors per hour* (Maximum likelihood), part (e); *Estimating a hidden timeout* (Maximum likelihood), part (c); *Estimating a hidden timeout* (Maximum likelihood), part (e).

Time between model failures · Sample C — 15 points

The time between failures of a deployed model is modeled as Exponential with rate λ (density $\lambda e^{-\lambda x}$). The observed gaps were 2, 4 and 6 hours.

Values you may use. $\sqrt{3} \approx 1.73$, $1/\sqrt{3} \approx 0.577$.

(a) **(4)** [Maximum likelihood] Derive the MLE of λ, check that it is a maximum, and compute it.

(b) **(3)** Find the Fisher information for one gap and the approximate standard error of $\hat{\lambda}$.

(c) **(3)** [Maximum likelihood] Find the method-of-moments estimator, and the MLE of the mean time between failures.

(d) **(3)** Is $\hat{\lambda} = 1/\bar{X}$ unbiased? Use the convexity of $1/x$ to decide the direction of any bias.

(e) **(2)** What large-sample property of the MLE justifies the interval $\hat{\lambda} \pm 1.96\,\mathrm{SE}$?

Squared error is a likelihood · Sample E — 15 points

A regression model's errors are assumed to be Normal(μ, σ^2).

(a) **(4)** [Maximum likelihood] For data $x_1, \dots, x_n$, write the log-likelihood and derive the MLEs of μ and σ^2.

(b) **(3)** [Maximum likelihood] The residuals on five validation examples are $1, -1, 3, -3, 0$. Find $\hat{\mu}$, $\hat{\sigma}^2$, and the unbiased variance estimate.

(c) **(3)** [Maximum likelihood] A network predicts $f(x_i)$, and $y_i = f(x_i) + \varepsilon_i$ with $\varepsilon_i \sim N(0, \sigma^2)$, where σ is fixed. Show that maximizing the likelihood is the same as minimizing $\sum (y_i - f(x_i))^2$.

(d) **(3)** Prove that for any estimator, $E[(\hat{\theta} - \theta)^2] = \mathrm{Var}(\hat{\theta}) + \big(E[\hat{\theta}] - \theta\big)^2$.

(e) **(2)** Given that $E[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2$, find the bias of $\hat{\sigma}^2$ when $n = 5$. Does it matter for large n?

Bayesian inference

Mon Nov 16 · presented by Brian

5 problems, 75 points.

How often does the jailbreak work? · Sample A **15 points**

A red team estimates the rate θ at which a jailbreak prompt succeeds. They use the prior $\theta \sim \text{Beta}(2, 8)$. In 10 attempts, the jailbreak succeeds 3 times.

(a) **(3)** Give the prior mean. Using the course convention, how many "pretend" successes and failures does this prior represent?

(b) **(3)** Show that the posterior is again a Beta distribution, and name its parameters.

(c) **(4)** Find the posterior mean, the MAP and the MLE. Explain their order.

(d) **(2)** What is the probability that the next attempt succeeds?

(e) **(3)** The 95% credible interval for θ is about $[0.09, 0.46]$. Write one sentence saying what it means. Then write one sentence saying what a 95% *confidence* interval means.

How much should the prior matter? · Sample B **15 points**

A model is tried on a new task. In the first 10 tries it succeeds 9 times. Let θ be its true success rate.

(a) **(4)** With a flat prior $\text{Beta}(1, 1)$, find the posterior, its mean and its MAP. Compare the MAP with the MLE.

(b) **(4)** A skeptical engineer uses $\text{Beta}(50, 50)$. Find the posterior and its mean. Using the course convention that $\text{Beta}(a, b)$ acts like $a - 1$ successes and $b - 1$ failures, how many pretend trials does this prior add, compared with the 10 real ones?

(c) **(4)** After 1,000 tries the model has 900 successes. Find both posterior means. What happened to the disagreement?

(d) **(3)** Show that the posterior mean $\dfrac{a + s}{a + b + n}$ is a weighted average of the prior mean $\dfrac{a}{a + b}$ and the sample proportion $\dfrac{s}{n}$, and give the weights.

Errors per day, the Bayesian way · Sample C **15 points**

The number of errors per day is $\text{Poisson}(\lambda)$. The prior on λ is $\text{Gamma}(\alpha = 2,\ \beta = 1)$, with density $\propto \lambda^{\alpha-1} e^{-\beta\lambda}$, mean α/β and mode $(\alpha - 1)/\beta$. Four days show 3, 4, 2 and 3 errors.

(a) **(4)** Show that the posterior is Gamma, and find its parameters.

(b) **(3)** Find the posterior mean, the MAP and the MLE.

(c) **(4)** Show that the posterior mean is a weighted average of the prior mean and $\bar{x}$, and give the weights. What do α and β act like?

(d) **(2)** The 95% credible interval is about $[1.5, 4.5]$. Interpret it in one sentence.

(e) **(2)** What happens to the posterior after 400 days of data?

One step of MCMC by hand · Sample D — 15 points

With a flat prior, the posterior for a jailbreak's success rate after 3 successes in 10 attempts is proportional to $\theta^3(1-\theta)^7$ for $0 \le \theta \le 1$ (and 0 elsewhere). A Metropolis sampler proposes a nearby value symmetrically, draws $U \sim \text{Uniform}(0,1)$, and moves if $U < \min(1, \text{ratio})$.

Values you may use. $(4/3)^3 \approx 2.37$, $(6/7)^7 \approx 0.34$.

(a) **(3)** Why does the sampler never need the posterior's normalizing constant?

(b) **(4)** The chain is at $\theta = 0.3$ and proposes 0.4. Find the acceptance probability. What happens if $U = 0.5$? If $U = 0.9$?

(c) **(2)** From 0.4, what is the acceptance probability for proposing 0.3? For proposing 1.2?

(d) **(3)** The exact posterior is Beta(4, 8). Find its mean and mode. What should the long-run average of the chain be close to?

(e) **(3)** Why does the sampler sometimes accept a worse value? Name two checks that the chain has converged.

A Bayesian quality score · Sample E — 15 points

A new model's true average human-preference score is μ. Based on earlier versions the prior is $\mu \sim N(70, 4^2)$. Four raters give scores averaging $\bar{x} = 78$, and each rating has noise SD $\sigma = 8$.

Values you may use. Posterior precision $= \frac{1}{\tau^2} + \frac{n}{\sigma^2}$, posterior mean $= \dfrac{\mu_0/\tau^2 + n\bar{x}/\sigma^2}{1/\tau^2 + n/\sigma^2}$; $\sqrt{8} \approx 2.83$, $z_{0.975} = 1.96$; here $\mu_0 = 70$ and $\tau = 4$ are the prior mean and SD.

(a) **(4)** Find the posterior mean and variance of μ.

(b) **(3)** Give a 95% credible interval and interpret it.

(c) **(3)** With 100 raters instead (same $\bar{x}$), find the posterior mean. What changed and why?

(d) **(3)** Go back to 4 raters. If the prior were nearly flat ($\tau \to \infty$), what would the posterior mean become, and how would the MAP compare with the MLE? What would a very small τ do?

(e) **(2)** How is the interval in (b) different in meaning from a 95% confidence interval?

Dig deeper (not on the final: for presenters and anyone who wants more)

Weight decay in one dimension (ridge regression) · Dig deeper — 10 points

Use the centered GPU data from *Does more GPU time buy a better score?* (Linear models): $S_{xy} = 8$ and $S_{xx} = 10$. Ridge regression chooses the slope b to minimize $\sum(y_i - bx_i)^2 + \lambda b^2$ (centered data).

(a) **(4)** Show that $\hat{b}_\lambda = \dfrac{S_{xy}}{S_{xx} + \lambda}$. Find it for $\lambda = 0$, 10 and ∞.

(b) **(4)** If the true slope is β and the noise has variance σ^2, then $E[\hat{b}_\lambda] = \beta\dfrac{S_{xx}}{S_{xx} + \lambda}$ and $\text{Var}(\hat{b}_\lambda) = \dfrac{\sigma^2 S_{xx}}{(S_{xx} + \lambda)^2}$. Describe what increasing λ does to the bias and the variance.

(c) **(2)** What prior on b makes ridge the MAP estimate, and how does λ relate to it?

Why lasso sets coefficients to exactly zero · Dig deeper **9 points**

With an orthonormal design, the lasso solves, separately for each coefficient, $\min_\beta \ \frac{1}{2}(z-\beta)^2 + \lambda|\beta|$, where z is the least-squares estimate.

(a) **(4)** Show that the solution is $\hat{\beta} = \text{sign}(z)\max(|z| - \lambda,\ 0)$ (soft-thresholding).

(b) **(3)** With $\lambda = 1$, apply it to $z = (3,\ 0.5,\ -1.2)$.

(c) **(2)** Compare with the ridge version (same $\lambda = 1$), $\min_\beta \frac{1}{2}(z-\beta)^2 + \frac{\lambda}{2}\beta^2$, whose solution is $\hat{\beta} = z/(1+\lambda)$. Which one selects features?

Hypothesis testing

Wed Nov 18 · presented by Shirley

5 problems, 75 points.

Also practice: *Reading cross-validation results* (Honest evaluation and causal claims), part (c).

Did the new assistant reduce thumbs-down ratings? · Sample A — 15 points

Each user is randomly shown the old or the new assistant. With the old one, 150 of 1,000 users give a thumbs-down (15%). With the new one, 120 of 1,000 do (12%).

> **Values you may use.** Pooled SE of the difference ≈ 0.015. $P(Z > 2) \approx 0.023$. $z_{0.975} = 1.96$, $z_{0.80} = 0.84$, and $(1.96 + 0.84)^2 = 7.84$.

(a) **(2)** State H_0 and H_1 in words and in symbols (two-sided).

(b) **(4)** Compute the test statistic and the two-sided p-value. What do you conclude at $\alpha = 0.05$?

(c) **(3)** Say exactly what this p-value is the probability of. Then name two things it is *not*.

(d) **(4)** In a related product whose thumbs-down rate is 20%, the team decides the smallest drop worth detecting is to 10%. How many users per group give 80% power at $\alpha = 0.05$? Use $n = \dfrac{(z_{0.975} + z_{0.80})^2 \left[p_1(1-p_1) + p_2(1-p_2)\right]}{(p_1 - p_2)^2}$. About how many would you need to detect a gap half as large (20% to 15%)?

(e) **(2)** In this product setting, describe a Type I error and a Type II error. Which is costlier?

A permutation test by hand · Sample B — 15 points

Three answers written with a new prompt template were rated 7, 9 and 8. Three written with the old template were rated 5, 6 and 7. Test whether the template matters.

(a) **(2)** Compute the observed difference in mean rating (new minus old).

(b) **(5)** Under H_0 the labels "new" and "old" are arbitrary, so every way of choosing 3 of the 6 ratings to call "new" is equally likely. How many ways are there? How many give a difference of at least 2? Find the one-sided and the two-sided p-values.

(c) **(3)** Explain in words what was shuffled, and why shuffling builds the null distribution.

(d) **(3)** Show that with 3 answers per group (all ratings different) the smallest possible two-sided p-value is 0.10. What does that imply?

(e) **(2)** Use (d) to explain Type II error and power for this design.

Testing a model in eight user segments · Sample C — 15 points

A team tests whether a model's error rate in each of 8 user segments differs from its overall rate. The sorted p-values are

$$0.001,\ 0.008,\ 0.012,\ 0.019,\ 0.04,\ 0.09,\ 0.30,\ 0.70.$$

> **Values you may use.** $0.95^{20} \approx 0.36$.

(a) **(3)** Apply Bonferroni at $\alpha = 0.05$. Which segments are flagged?

(b) **(5)** Apply Benjamini–Hochberg at $q = 0.05$. Show the threshold for each rank and say which segments are flagged.

(c) **(3)** Suppose instead the team tests 20 segments, the model truly behaves the same in all 20, and each segment is tested independently at level 0.05. How many false alarms do you expect? What is the probability of at least one?

(d) **(2)** Define the family-wise error rate and the false discovery rate.

(e) **(2)** When would you use Bonferroni, and when BH?

Planning a latency A/B test · Sample D — 15 points

A team will test whether a new serving stack lowers response time. They randomize requests to the old and new stack. The response time has SD $\sigma = 20$ ms.

Values you may use. $n = \dfrac{2(z_{0.975} + z_{0.80})^2 \sigma^2}{\delta^2}$ per group, where $(1.96 + 0.84)^2 = 7.84$.

(a) **(4)** How many requests per group detect a true drop of $\delta = 5$ ms with 80% power at $\alpha = 0.05$ (two-sided)?

(b) **(2)** About how many are needed to detect a 2.5 ms drop?

(c) **(3)** Define power. Name the four quantities that are linked, so that fixing three sets the fourth.

(d) **(4)** True or false? Give a reason for each. (i) $p = 0.03$ means a 3% chance that H_0 is true. (ii) $p = 0.2$ proves the new stack makes no difference. (iii) With 10 million requests per group, a 0.1 ms improvement can give $p < 0.001$. (iv) α is the Type I error rate when H_0 is true.

(e) **(2)** An engineer checks the p-value every hour and stops as soon as $p < 0.05$. What goes wrong?

Does the model meet its advertised accuracy? · Sample E — 15 points

A vendor claims 80% accuracy. On 400 independent test items, the model gets 300 right.

Values you may use. $P(Z > 2.5) \approx 0.006$, $\sqrt{0.75 \times 0.25/400} \approx 0.0217$, $z_{0.975} = 1.96$.

(a) **(4)** Test $H_0 : p = 0.8$ against $H_1 : p \neq 0.8$. Give z and the p-value.

(b) **(3)** Build a 95% confidence interval for p. Does it contain 0.8?

(c) **(3)** Explain why (a) and (b) agree.

(d) **(3)** Another model scores 0.799 on 1,000,000 items. Test the same claim. What does this show about p-values and effect size?

(e) **(2)** The vendor ran 20 benchmarks and advertised the one where the model looked best. Why should you discount that result?

Dig deeper (not on the final: for presenters and anyone who wants more)

The smallest effect an A/B test can see · Dig deeper — 5 points

Values you may use. $\sqrt{0.005} \approx 0.0707$.

(a) **(3)** A latency A/B test (two-sided $\alpha = 0.05$, 80% power) has a per-request SD of $\sigma = 20$ ms and 400 requests per arm. Find the minimum detectable effect, MDE $= (z_{0.975} + z_{0.80})\,\sigma\sqrt{2/n}$, using $1.96 + 0.84 = 2.8$.

(b) **(2)** How many requests per arm would halve the MDE?

Linear models

Fri Nov 20 · presented by Ronny

5 problems, 75 points.

Also practice: *When the total points the wrong way* (Probability and Bayes' rule), part (b); *Two coordinates of an embedding* (The covariance matrix and the multivariate normal), part (e); *What PCA sees in a feature set* (PCA), part (d).

Does more GPU time buy a better score? · Sample A **15 points**

Five fine-tuning runs: x = GPU hours beyond a 100-hour base run (in hundreds) and y = benchmark score.

x	0	1	2	3	4
y	1	3	2	5	4

(a) **(4)** Find the least-squares line $\hat{y} = a + bx$. Show $\bar{x}$, $\bar{y}$, $S_{xy} = \sum(x - \bar{x})(y - \bar{y})$ and S_{xx}.

(b) **(3)** Find the residuals. Check that they sum to 0, and find the residual sum of squares.

(c) **(3)** Find R^2 and the correlation r. How are they related here?

(d) **(3)** Interpret the slope in one sentence. Then explain why it is not the causal effect of GPU time, naming a plausible confounder.

(e) **(2)** Prove that the residuals sum to 0 whenever the model has an intercept.

Least squares as a projection · Sample B **15 points**

Fit $y = \beta_0 + \beta_1 x$ to three runs of a learning-rate sweep: $x = (-1, 0, 1)$ (coded learning rate) and $y = (1, 2, 6)$ (validation gain). Let X be the 3×2 design matrix with a column of ones and the column x.

(a) **(4)** Starting from the goal of minimizing $\|y - X\beta\|^2$, derive the normal equations $X^T X\beta = X^T y$.

(b) **(4)** Compute $X^T X$ and $X^T y$, and solve for $\hat{\beta}$.

(c) **(3)** Find the fitted values and residuals. Show that the residual vector is orthogonal to both columns of X.

(d) **(2)** Describe $\hat{y}$ geometrically.

(e) **(2)** Adding a column of pure random noise to X never increases the training SSE. Explain why, using (d). What does this say about judging a model by training R^2?

Does the AI tutor raise exam scores? · Sample C **15 points**

Suppose the truth is score $= 2\,(\text{tutor use}) + 5\,(\text{GPA}) + \varepsilon$, where tutor use $= 0.5\,(\text{GPA}) + u$. Here GPA, u and ε are independent, with $\text{Var}(\text{GPA}) = 1$ and $\text{Var}(u) = 0.75$ (GPA and tutor use are in standardized units).

(a) **(3)** Find Var(tutor use) and Cov(tutor use, GPA).

(b) **(5)** An analyst regresses score on tutor use alone. Find the slope they get:

$$\frac{\text{Cov}(\text{score}, \text{use})}{\text{Var}(\text{use})}.$$

(c) **(3)** Explain the extra 2.5, and what the coefficient 2 means in the full model.

(d) **(2)** [Honest evaluation and causal claims] What study design would estimate the causal effect without measuring GPA at all?

(e) **(2)** [Honest evaluation and causal claims] Draw the causal diagram, with arrows.

Diagnosing a cost model · Sample D — 15 points

A team regresses the inference cost of a deployed model on model size, batch size and prompt length.

Values you may use. $\sqrt{10} \approx 3.16$, $1/0.0975 \approx 10.3$.

(a) **(4)** For each residual-plot pattern, say what it tells you: (i) a U-shaped curve; (ii) a funnel that widens to the right; (iii) a drift when residuals are plotted in time order; (iv) one isolated point far to the right with a large residual.

(b) **(4)** The variance inflation factor is $\text{VIF}_j = 1/(1 - R_j^2)$, where R_j^2 comes from regressing feature j on the other features. Find the VIF when $R_j^2 = 0.9$, and say what it does to that coefficient's SE. Then find the VIF for two features with correlation 0.95, when they are the only two features in the model.

(c) **(3)** Model size and batch size have correlation 0.95. Why does this barely hurt cost *prediction* but badly hurt *interpretation* of each coefficient?

(d) **(2)** Other than constant variance, which assumption matters most for the standard errors? Give an AI example that breaks it.

(e) **(2)** What do you do about the funnel in (a)(ii)?

Is the resume screener biased? · Sample E — 15 points

An AI screener's interview recommendations, by department and applicant gender:

	Men	**Women**
Department E	160 of 200 recommended	45 of 50 recommended
Department H	10 of 50 recommended	60 of 200 recommended

(a) **(3)** Find each group's recommendation rate in each department and overall.

(b) **(3)** Explain how women can have higher rates in both departments but a lower overall rate.

(c) **(4)** Suppose each gender applied half to E and half to H. Find each group's overall rate, and name the rule you used.

(d) **(3)** [Logistic regression] A logistic regression of "recommended" on gender alone, and another on gender and department: what sign does the coefficient for *woman* have in each? Explain.

(e) **(2)** [Honest evaluation and causal claims] Does (d) prove the screener treats equally qualified applicants fairly? What else would you need?

Logistic regression

Mon Nov 23 · presented by Ty'Kuan

5 problems, 75 points.

Also practice: *Softmax turns scores into a distribution* (Random variables and distributions), part (e); *Is the resume screener biased?* (Linear models), part (d).

Reading a fraud model's coefficients · Sample A — 15 points

A logistic regression flags fraud with $\log \dfrac{p}{1-p} = -2 + 0.7\,(\text{prior flags}) + 1.1\,(\text{new account})$, where *new account* is 1 or 0.

Values you may use. $e^{0.7} \approx 2.0$, $e^{1.1} \approx 3.0$, $e^{-0.5} \approx 0.61$, $e^{2} \approx 7.4$.

(a) **(3)** Give the odds ratio for one extra prior flag, and write it as a sentence.

(b) **(4)** Find p for a new account with 2 prior flags.

(c) **(3)** Find p for an old account with no prior flags.

(d) **(3)** Show that $\log \dfrac{p}{1-p} = z$ is the same statement as $p = \dfrac{1}{1+e^{-z}}$.

(e) **(2)** One extra flag always doubles the odds. Explain why the change in the *probability* depends on where you start, using $\dfrac{dp}{dz} = p(1-p)$.

Why classifiers train on cross-entropy · Sample B — 15 points

A classifier outputs $p = P(y = 1 \mid x)$ for each example.

Values you may use. $\ln 0.9 \approx -0.105$, $\ln 0.8 \approx -0.223$, $\ln 0.6 \approx -0.511$, $\ln 0.01 \approx -4.61$.

(a) **(4)** For one example with label $y \in \{0, 1\}$, write the Bernoulli likelihood. Show that its negative log is the cross-entropy loss $-[\,y \ln p + (1-y)\ln(1-p)\,]$.

(b) **(4)** Find the mean cross-entropy over three examples: $(y = 1, p = 0.9)$, $(y = 0, p = 0.2)$ and $(y = 1, p = 0.6)$.

(c) **(2)** Find the loss for $y = 1$ with $p = 0.01$. What does this show?

(d) **(3)** Let $p = \sigma(z) = 1/(1+e^{-z})$. Using $\sigma'(z) = \sigma(z)(1-\sigma(z))$, show that $\dfrac{\partial\,\text{loss}}{\partial z} = p - y$.

(e) **(2)** Explain why minimizing the average cross-entropy over a training set is maximum likelihood.

From Naive Bayes to logistic regression · Sample C — 15 points

A moderation model labels messages *toxic* or *ok* using two yes/no features: x_1 = a slur is detected, and x_2 = the message is written in all caps. Assume the features are independent given the class.

	$P(\cdot = 1 \mid \text{toxic})$	$P(\cdot = 1 \mid \text{ok})$
x_1 (slur)	0.6	0.2
x_2 (all caps)	0.5	0.25

$P(\text{toxic}) = 0.2$

(a) **(4)** Show that the log posterior odds are a linear function of x_1 and x_2, which is the logistic regression form.

(b) **(4)** Find $P(\text{toxic})$ for a message with $x = (1, 1)$, and for one with $x = (1, 0)$.

(c) **(3)** Find w_0, w_1 and w_2 as logarithms. Check them against (b).

(d) **(2)** Naive Bayes and logistic regression share this form. How do they differ in how they find the weights?

(e) **(2)** Show that two-class softmax equals the sigmoid of the score difference.

Thresholds and odds ratios · Sample D — 15 points

(a) **(4)** A content filter outputs $p = P(\text{harmful})$. Missing a harmful post costs $C_{FN} = 9$; wrongly blocking a safe post costs $C_{FP} = 1$. Show that you should block exactly when $p > \dfrac{C_{FP}}{C_{FP} + C_{FN}}$, and find the threshold.

(b) **(4)** Hallucinations in an audit of two model versions: version 1 had 60 hallucinated and 240 fine answers; version 2 had 30 and 270. Find each version's odds of hallucinating and the odds ratio (version 2 to version 1).

(c) **(3)** A logistic regression with one 0/1 feature ($v_2 = 1$ for version 2) is fit to these data. What are its intercept and slope?

(d) **(2)** Find the ratio of the hallucination *probabilities*. Why does it differ from the odds ratio?

(e) **(2)** Write one sentence reporting the result for a product manager.

Who clicks "regenerate"? · Sample E — 15 points

A logistic regression predicts whether a user clicks *regenerate* (a sign of a bad answer). Device is coded with desktop as the baseline: $\log \dfrac{p}{1-p} = -1 + 0.4\,(\text{phone}) - 0.2\,(\text{tablet})$.

Values you may use. $e \approx 2.72$, $e^{0.6} \approx 1.82$, $e^{0.4} \approx 1.49$, $e^{-0.2} \approx 0.82$.

(a) **(3)** Write the likelihood of a logistic regression for data (x_i, y_i), $i = 1, \ldots, n$. Why is there no closed-form MLE?

(b) **(4)** For one numeric feature with coefficient $\beta = 0.8$, $\dfrac{dp}{dx} = \beta p(1-p)$. Find this effect at $p = 0.5$ and at $p = 0.9$. Explain the difference.

(c) **(3)** Interpret the phone and tablet coefficients as odds ratios. Why is there no "desktop" term?

(d) **(3)** Find p for a desktop user and for a phone user.

(e) **(2)** An analyst refits the same model with *phone* as the baseline instead of desktop. Give the new intercept and the coefficients for desktop and tablet.

Dig deeper (not on the final: for presenters and anyone who wants more)

Entropy, cross-entropy and KL divergence · Dig deeper — 8 points

Use base-2 logarithms. Let $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ be the true next-token distribution, $q = (\frac{1}{4}, \frac{1}{4}, \frac{1}{2})$ a model's distribution, and $r = (\frac{1}{2}, \frac{1}{2}, 0)$ another model's.

(a) **(3)** Find the entropy $H(p)$, the cross-entropy $H(p, q) = -\sum p_k \log_2 q_k$, and $D_{KL}(p \,\|\, q)$. Check that $H(p, q) = H(p) + D_{KL}(p \,\|\, q)$.

(b) **(3)** Find $D_{KL}(r \,\|\, p)$ and $D_{KL}(p \,\|\, r)$. What does this show?

(c) **(2)** Why does minimizing cross-entropy in training mean minimizing KL divergence?

The softmax gradient · Dig deeper **6 points**

A model has logits $z = (\ln 9, \ln 4, 0)$, so softmax gives $p = (\frac{9}{14}, \frac{4}{14}, \frac{1}{14})$. The true class is the second one, so the one-hot label is $y = (0, 1, 0)$. The loss is $L = -\ln p_2$.

(a) **(4)** Show that $\dfrac{\partial L}{\partial z_k} = p_k - y_k$, and compute the gradient.

(b) **(2)** One step of gradient descent moves z by $-\eta\times$ the gradient. Which logits rise and which fall? Is that sensible?

Honest evaluation and causal claims

Wed Nov 25 · presented by Brian

5 problems, 75 points.

Also practice: *Does the AI tutor raise exam scores?* (Linear models), part (d); *Does the AI tutor raise exam scores?* (Linear models), part (e); *The interview screen that invents a pattern* (Probability and Bayes' rule), part (e); *Is the resume screener biased?* (Linear models), part (e).

Evaluating a fraud model honestly · Sample A — 15 points

A test set has 10,000 transactions, 100 of them fraud. The model flags 480 transactions: 80 are fraud and 400 are legitimate.

(a) **(4)** Write the confusion matrix: TP, FN, FP, TN.

(b) **(4)** Find precision, recall, F_1 and accuracy. Compare accuracy with a model that calls everything legitimate.

(c) **(3)** Which numbers would you report to the fraud team, and why? What curve would you show instead of an ROC curve?

(d) **(4)** A colleague's model reaches 99% recall using a feature called *chargeback filed*. Many customers also appear several times in both the training and test sets. Name both problems and say how you would fix each.

Leaks that make a model look better than it is · Sample B — 15 points

(a) **(5)** For each pipeline, say whether information leaks from the test data, and give the fix.

i. Features are standardized using the mean and SD of the *whole* dataset, then the data are split.

ii. A clinic's patients each have several visits, and visits are split at random.

iii. Next week's GPU demand is forecast, and days are split by random 5-fold cross-validation.

iv. The 20 features most correlated with the outcome are chosen using all rows, and then the model is cross-validated.

v. 50 hyperparameter settings are scored on the test set, and the best score is reported.

(b) **(4)** The clinic data have 1,000 visits from 200 patients. In group 5-fold cross-validation, how many patients are in each fold? Why must all of a patient's visits stay in one fold?

(c) **(3)** Explain why reporting the best of 50 test scores is biased upward, even if no single score is biased.

(d) **(3)** Describe a rolling-origin (time-based) evaluation for the GPU forecast.

Can we trust the model's probabilities? · Sample C **15 points**

A support-ticket triage model predicts the probability that a ticket is urgent. On 500 test tickets the predictions fall into three groups:

Predicted probability	Tickets	Actually urgent
0.1	200	40
0.5	100	50
0.9	200	150

(a) **(3)** Find the observed urgent rate in each group.

(b) **(4)** The expected calibration error is the ticket-weighted average of $|\text{predicted} - \text{observed}|$. Compute it.

(c) **(3)** Is the model overconfident or underconfident? Which way should the corrected 0.1 and 0.9 move?

(d) **(3)** The support team staffs from the sum of the predicted probabilities. Compute the predicted number of urgent tickets and the actual number. Why does calibration matter for this use?

(e) **(2)** Can a model have high accuracy and poor calibration? Give an example.

Reading cross-validation results · Sample D **15 points**

Five-fold cross-validation accuracies for model A were 0.80, 0.84, 0.78, 0.82 and 0.76.

Values you may use. $\sqrt{0.001} \approx 0.032$, $\sqrt{5} \approx 2.24$.

(a) **(4)** Find the mean accuracy and the sample standard deviation (divide by $n - 1$).

(b) **(3)** A naive SE of the mean is $\text{SD}/\sqrt{5}$. Compute it. Why is it probably too small?

(c) **(4)** [Hypothesis testing] Model B, on the *same* folds, scored 0.82, 0.85, 0.80, 0.83 and 0.78. Find the fold-by-fold differences. If the models were equally good, each difference would be positive with probability $\frac{1}{2}$. Find the probability that all five are positive, and say what you conclude.

(d) **(2)** The team tried 20 configurations and reports the best cross-validation score as the final accuracy. What is wrong?

(e) **(2)** Why use 5-fold cross-validation rather than a single 80/20 split?

Fair on paper, unequal in practice · Sample E **15 points**

A toxicity classifier has the same true-positive rate (0.9) and false-positive rate (0.1) for posts written in two dialects. In dialect group 1, 10% of posts are toxic; in group 2, 30% are.

(a) **(4)** For 1,000 posts from each group, find the true and false positives and the precision.

(b) **(3)** Explain why equal error rates give unequal precision. Can a classifier make error rates *and* precision equal across groups when base rates differ?

(c) **(3)** Which metric matters to a writer in group 1? To the moderation team?

(d) **(3)** A dashboard shows that users of the AI writing assistant are flagged less often. A manager concludes the assistant *causes* better behavior. What is wrong, and what study would answer the question?

(e) **(2)** Name one confounder for (d) and say which way it biases the dashboard's comparison.

Dig deeper (not on the final: for presenters and anyone who wants more)

Fixing overconfidence with temperature scaling · Dig deeper **7 points**

In *Can we trust the model's probabilities?* (Honest evaluation), the model's 0.9 predictions came true 75% of the time. Temperature scaling replaces $p = \sigma(z)$ with $\sigma(z/T)$.

(a) **(3)** A prediction of 0.9 has logit $z = \ln 9$. Find T so that it becomes 0.75.

(b) **(2)** With $T = 2$, what does a 0.1 prediction become? Compare with the observed 0.2.

(c) **(2)** Does temperature scaling change which tickets are ranked most urgent? Why fit T on held-out data?

Watching for data drift · Dig deeper **6 points**

(a) **(3)** A feature's mean was 10.0 last month and 10.3 this month, and the standard error of the difference is 0.1. Test for drift at $\alpha = 0.05$.

(b) **(3)** A dashboard runs this test daily on 50 features at $\alpha = 0.05$. How many false alarms should you expect each day if nothing drifts? What would you change?